

## chap 8 Advanced Counting Techniques

- 8.1 Recurrence Relations
- 8.2 Solving Recurrence Relations
  - Linear homogeneous recurrence relations with constant coefficients of degree  $k$   
( $k$ 阶齐次常系数线性递推关系)
- 8.3 Divide-and-Conquer Algorithms and Recurrence Relations
- 8.4 Generating Functions

- 8.1-14. a) Find a recurrence relation for the number of ternary strings of length  $n$  that contain two consecutive 0s.
  - b) What are the initial conditions?
  - c) How many ternary strings of length six contain two consecutive 0s?
- 
- $a_n = 2a_{n-1} + 2a_{n-2} + 3^{n-2}$
  - $a_0 = 0, a_1 = 0$
  - $a_2 = 1, a_3 = 5, a_4 = 21, a_5 = 79, a_6 = 281$

## 8.2

46. Suppose that there are two goats on an island initially. The number of goats on the island doubles every year by natural reproduction, and some goats are either added or removed each year.

a) Construct a recurrence relation for the number of goats on the island at the start of the  $n$ th year, assuming that during each year an extra 100 goats are put on the island.

b) Solve the recurrence relation from part (a) to find the number of goats on the island at the start of the  $n$ th year.

c) Construct a recurrence relation for the number of goats on the island at the start of the  $n$ th year, assuming that  $n$  goats are removed during the  $n$ th year for each  $n \geq 3$ .

d) Solve the recurrence relation in part (c) for the number of goats on the island at the start of the  $n$ th year.

$$a) a_n = 2a_{n-1} + 100$$

$$b) a_n^{(h)} = \alpha 2^n$$

particular solution  $a_n = c$

$$c = 2c + 100, c = -100$$

$$a_n = \alpha 2^n - 100$$

$$a_1 = 2 = \alpha 2 - 100 \quad \alpha = 51$$

$$a_n = 51 \cdot 2^n - 100$$

$$c) a_1 = 2, a_2 = 4,$$

$$a_n = 2a_{n-1} - n \quad n \geq 3 \text{ (assume remove after doubling)}$$

b)

particular solution  $a_n = cn + d$

$$cn + d = 2cn - 2c + 2d - n, c = 1, d = 2$$

$$a_n = \alpha 2^n + n + 2 \quad a_3 = 5 = 8\alpha + 5$$

$$\alpha = 0$$

$$a_n = n + 2 \quad n \geq 3$$

**36.** Use generating functions to solve the recurrence relation  $a_k = 3a_{k-1} + 4^{k-1}$  with the initial condition  $a_0 = 1$ .

Let  $G(x) = \sum_{k=0}^{\infty} a_k x^k$ . Then  $xG(x) = \sum_{k=0}^{\infty} a_k x^{k+1} = \sum_{k=1}^{\infty} a_{k-1} x^k$  (by changing the name of the variable from  $k$  to  $k+1$ ). Thus

$$\begin{aligned} G(x) - 3xG(x) &= \sum_{k=0}^{\infty} a_k x^k - \sum_{k=1}^{\infty} 3a_{k-1} x^k = a_0 + \sum_{k=1}^{\infty} (a_k - 3a_{k-1}) x^k = 1 + \sum_{k=1}^{\infty} 4^{k-1} x^k \\ &= 1 + x \sum_{k=1}^{\infty} 4^{k-1} x^{k-1} = 1 + x \sum_{k=0}^{\infty} 4^k x^k = 1 + x \cdot \frac{1}{1-4x} = \frac{1-3x}{1-4x}. \end{aligned}$$

Thus  $G(x)(1-3x) = (1-3x)/(1-4x)$ , so  $G(x) = 1/(1-4x)$ . Therefore  $a_k = 4^k$ , from Table 1.

常见问题：错误率较高

## Exercise

- Find the number of bit strings of length eight that contain a pair of consecutive 0s.
- Find the solution to the recurrence relation  $a_n = 6a_{n-1} - 5a_{n-2}$  for  $n \geq 2$  with initial conditions  $a_0 = 1, a_1 = 0$ .
- Find the solution to the recurrence relation  $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$  with the initial condition  $a_0 = 1, a_1 = -2$ , and  $a_2 = -1$ .
- Find all solutions to  $a_n = 3a_{n-1} + 2n$ . Which solution has  $a_1 = 3$ ?
- a) What is the generating function for  $\{a_k\}$ , where  $a_k$  is the number of solutions of  $x_1 + x_2 + x_3 + x_4 = k$  when  $x_1, x_2, x_3, x_4$  are integers with  $x_1 \geq 3, 1 \leq x_2 \leq 5, 0 \leq x_3 \leq 4$ , and  $x_4 \geq 1$ ?
- b) Use your answer to part (a) to find  $a_7$ .
- Set up a generating function and use it to find the number of ways in which ten identical boxes can be given to three children, if each child gets at least two boxes.
- Use generating functions to solve  $a_n = 7a_{n-1} - 10a_{n-2}$ ,  $a_0 = 1, a_1 = 1$ .

# Types of Graphs

- Simple graph
- Multigraph
- Pseudograph
- Simple directed graph
- Directed multigraph
- Mixed graph

# Types of Graphs: Summary

- Summary of the book's definitions.
- Keep in mind this terminology is not fully standardized across different authors...

| <b>TABLE 1 Graph Terminology.</b> |                         |                                |                       |
|-----------------------------------|-------------------------|--------------------------------|-----------------------|
| <i>Type</i>                       | <i>Edges</i>            | <i>Multiple Edges Allowed?</i> | <i>Loops Allowed?</i> |
| Simple graph                      | Undirected              | No                             | No                    |
| Multigraph                        | Undirected              | Yes                            | No                    |
| Pseudograph                       | Undirected              | Yes                            | Yes                   |
| Simple directed graph             | Directed                | No                             | No                    |
| Directed multigraph               | Directed                | Yes                            | Yes                   |
| Mixed graph                       | Directed and undirected | Yes                            | Yes                   |

## §10.2 Graph Terminology

You need to learn the following terms:

- *Adjacent, connects, endpoints, degree, initial, terminal, in-degree, out-degree, complete, cycles, wheels, n-cubes, bipartite, subgraph, union.*

Special cases of undirected graph structures:

- Complete graphs  $K_n$
- Cycles  $C_n$
- Wheels  $W_n$
- n-Cubes  $Q_n$
- Bipartite graphs
- Complete bipartite graphs  $K_{m,n}$



## §10.3: Graph Representations & Isomorphism

- Graph representations:
  - Adjacency lists.
  - Adjacency matrices.
  - Incidence matrices.
- Graph isomorphism:
  - Two graphs are isomorphic iff they are identical except for their node names.
- In an undirected graph, a *path of length  $n$  from  $u$  to  $v$*  is a sequence of adjacent edges going from vertex  $u$  to vertex  $v$ .
- A path is a *circuit* if  $u=v$ .
- A path *pass through* the vertices *or traverses* the edges.
- A path is *simple* if it contains no edge more than once.

# Connectedness

- An undirected graph is *connected* if there is a path between every pair of distinct vertices in the graph.
- A *cut vertex* or *cut edge (bridge)* separates 1 connected component into 2 if removed.
- A *Connected component* of a graph  $G$  is a connected subgraph of  $G$  that is not a proper subgraph of another connected subgraph of  $G$ .
- Vertex Connectivity
  - Nonseparable graphs
    - Connected graphs without cut vertices are called nonseparable graphs
  - Vertex cut
    - A subset  $V'$  of the vertex set  $V$  of  $G = (V, E)$  is a vertex cut, or separating set, if  $G - V'$  is disconnected.

- the **vertex connectivity** of a noncomplete graph  $G$ , denoted by  $\kappa(G)$ , as the minimum number of vertices in a vertex cut.
  - $0 \leq \kappa(G) \leq n - 1$  if  $G$  has  $n$  vertices
  - $\kappa(K_n) = n - 1$
  - $\kappa(G) = 0$  if and only if  $G$  is disconnected or  $G = K_1$
  - A graph is  $k$ -connected (or  $k$ -vertex-connected), if  $\kappa(G) \geq k$ .
- Edge Connectivity
  - Edge cut
    - A set of edges  $E$  is called an **edge cut** of  $G$  if the subgraph  $G - E$  is disconnected.
  - Edge connectivity
    - The **edge connectivity** of a graph  $G$ , denoted by  $\lambda(G)$ , is the minimum number of edges in an edge cut of  $G$ .
    - $\lambda(G) = 0$  if  $G$  is not connected or  $G$  has a single vertex.
    - If  $G$  is a graph with  $n$  vertices, then  $0 \leq \lambda(G) \leq n - 1$ .
    - $\lambda(G) = n - 1$  where  $G$  is a graph with  $n$  vertices if and only if  $G = K_n$ .

# Directed Connectedness

- A directed graph is *strongly connected* if there is a path from  $a$  to  $b$  and from  $b$  to  $a$  whenever  $a$  and  $b$  are vertices in the graph.
- It is *weakly connected* if there is a path between every two vertices in the underlying *undirected* graph.
- Note *strongly* implies *weakly* but not vice-versa.

## Counting Paths by Adjacency Matrices

Let  $\mathbf{A}$  be the adjacency matrix of graph  $G$ .

The number of paths of length  $k$  from  $v_i$  to  $v_j$  is equal to  $(\mathbf{A}^k)_{i,j}$ .

The notation  $(\mathbf{M})_{i,j}$  denotes  $m_{i,j}$  where  $[m_{i,j}] = \mathbf{M}$ .

## Eulerian graph (欧拉图)

- An ***Euler circuit*** in a graph  $G$  is a simple circuit containing every edge of  $G$ .
- An ***Euler path*** in  $G$  is a simple path containing every edge of  $G$ .
- A walk in a graph is called an ***Euler tour*** if it starts and ends in the same place and uses each edge exactly once.
- A walk in a graph is called an ***Euler trail*** if it uses each edge exactly once.
- If a graph has an Euler tour, it is said to be an ***Eulerian graph***.

## Hamiltonian Graph (哈密顿图)

- A graph has a *Hamiltonian tour* if there is a tour that visits every vertex exactly once (and returns to its starting point).
- A graph with a Hamiltonian tour is called a *Hamiltonian graph*.
- A *Hamiltonian path* is a path that contains each vertex exactly once.
- A *Hamilton circuit* is a circuit that traverses each vertex in  $G$  exactly once.
- A *Hamilton path* is a path that traverses each vertex in  $G$  exactly once

# Shortest Path Problem

- There are two types of such problems
  - Determining the shortest path from a vertex to an assigned vertex.
  - Determining the shortest path of any two vertices in the graph.

## ALGORITHM 1 Dijkstra's Algorithm.

```
procedure Dijkstra( $G$ : weighted connected simple graph, with  
    all weights positive)  
    { $G$  has vertices  $a = v_0, v_1, \dots, v_n = z$  and lengths  $w(v_i, v_j)$   
    where  $w(v_i, v_j) = \infty$  if  $\{v_i, v_j\}$  is not an edge in  $G$ }  
    for  $i := 1$  to  $n$   
         $L(v_i) := \infty$   
     $L(a) := 0$   
     $S := \emptyset$   
    {the labels are now initialized so that the label of  $a$  is 0 and all  
    other labels are  $\infty$ , and  $S$  is the empty set}  
    while  $z \notin S$   
         $u :=$  a vertex not in  $S$  with  $L(u)$  minimal  
         $S := S \cup \{u\}$   
        for all vertices  $v$  not in  $S$   
            if  $L(u) + w(u, v) < L(v)$  then  $L(v) := L(u) + w(u, v)$   
            {this adds a vertex to  $S$  with minimal label and updates the  
            labels of vertices not in  $S$ }  
    return  $L(z)$  { $L(z)$  = length of a shortest path from  $a$  to  $z$ }
```

## ALGORITHM 2 Floyd's Algorithm.

```
procedure Floyd( $G$ : weighted simple graph)  
    { $G$  has vertices  $v_1, v_2, \dots, v_n$  and weights  $w(v_i, v_j)$   
    with  $w(v_i, v_j) = \infty$  if  $\{v_i, v_j\}$  is not an edge}  
    for  $i := 1$  to  $n$   
        for  $j := 1$  to  $n$   
             $d(v_i, v_j) := w(v_i, v_j)$   
    for  $i := 1$  to  $n$   
        for  $j := 1$  to  $n$   
            for  $k := 1$  to  $n$   
                if  $d(v_j, v_i) + d(v_i, v_k) < d(v_j, v_k)$   
                    then  $d(v_j, v_k) := d(v_j, v_i) + d(v_i, v_k)$   
    return  $[d(v_i, v_j)]$  { $d(v_i, v_j)$  is the length of a shortest  
    path between  $v_i$  and  $v_j$  for  $1 \leq i \leq n, 1 \leq j \leq n$ }
```

# Planar Graphs – 平面图

- A graph is called *planar* if it can be drawn in the plane in such a way that no two edges cross.
- Example of a planar graph: The clique on 4 nodes.
- Euler Theorem
  - Let  $G$  be a connected planar simple graph with  $e$  edges and  $v$  vertices. Let  $r$  be the number of regions in a planar representation of  $G$ . Then  $r = e - v + 2$ .
- Every (simple)  $n$ -node planar graph  $G$  has at most  $3n-6$  edges.
- If  $G$  is a connected planar simple graph, then  $G$  has a vertex of degree not exceeding five.
- If a connected planar simple graph has  $e$  edges and  $v$  vertices with  $v \geq 3$  and no circuit of length three, then  $e \leq 2v-4$
- Kuratowski's Theorem

A graph is nonplanar if and only if it contains a subgraph homeomorphic to  $K_{3,3}$  or  $K_5$ .



# Coloring graphs

- Suppose that  $G = (V, E, \gamma)$  is a graph with no multiple edges, and  $C = \{c_1, c_2, \dots, c_n\}$  is any set of  $n$  “colors”.
- Any function  $f: V \rightarrow C$  is called *a coloring of the graph  $G$  using  $n$  colors* (or using the colors of  $C$ ).
  - For each vertex  $v$ ,  $f(v)$  is the color of  $v$ .
- A coloring is *proper* (合适的) if any two adjacent vertices  $v$  and  $u$  have different colors.
- The smallest number of colors needed to produce a proper coloring of a graph  $G$  is called the *chromatic number of  $G$*  ( $G$ 的着色数), denoted by  $\chi(G)$ .
- Computing the total number of different proper colorings of a graph  $G$  using a set  $\{c_1, c_2, \dots, c_n\}$  of colors.
- If  $G$  is a graph and  $n \geq 0$  is an integer, let  $P_G(n)$  be the number of ways to color  $G$  properly using  $n$  or fewer colors.
- $P_G$  is called the *chromatic polynomial* (着色多项式) of  $G$ .

## Theorem 1

- If  $G$  is a disconnected graph with components  $G_1, G_2, \dots, G_m$ , then  $P_G(x)$  is the product of the chromatic polynomials for each component
  - $P_G(x) = P_{G_1}(x) P_{G_2}(x) \dots P_{G_m}(x)$

## Theorem 2

Let  $G = (V, E, \gamma)$  be a graph with no multiple edges, and let  $e \in E$ , say  $e = \{a, b\}$ .

$G_e$  : subgraph of  $G$  obtained by deleting  $e$ ,

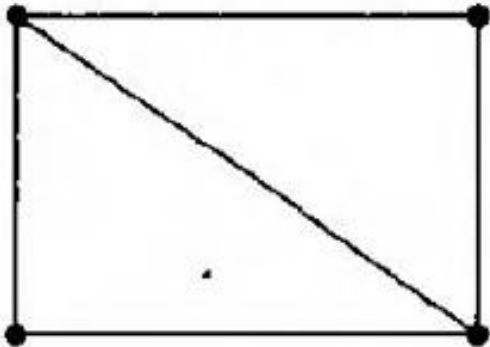
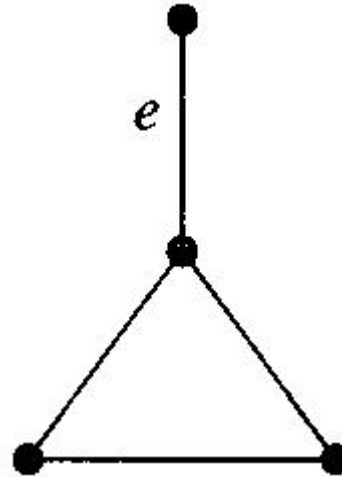
$G^e$  : the quotient graph of  $G$  obtained by merging the end points of  $e$ .

Then with  $x$  colors:

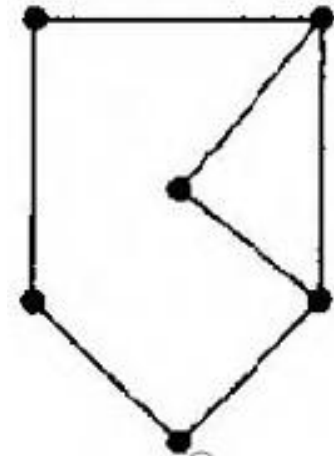
$$P_G(x) = P_{G_e}(x) - P_{G^e}(x).$$

## Example

- By Theorem 1
  - $P_{G_e}(x) = x(x(x-1)(x-2)) = x^2(x-1)(x-2)$
  - $P_{G^e}(x) = x(x-1)(x-2)$
- By Theorem 2,
  - $P_G(x) = x^2(x-1)(x-2) - x(x-1)(x-2)$   
 $= x(x-1)^2(x-2)$
- So,  $\chi(G) = 3$



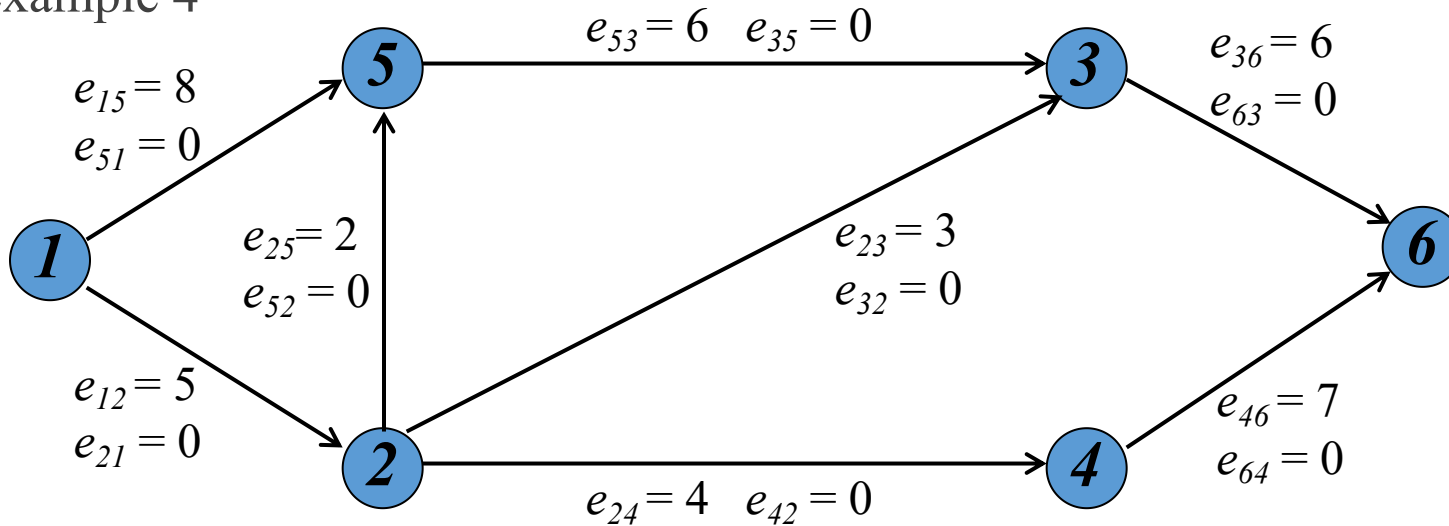
$$x(x-1)(x-2)^2$$



$$x(x-1)(x-2)^2(x^2-2x+2)$$

# transport network

- Application of digraph
- Concept
  - Capacity (容量)
  - Transport network (传输网)
  - Flow (流量)
  - Virtual flow (虚流量)
  - Maximal flow (最大流量)
- Labeling algorithm (标记算法)
- Example 4



**44.** Determine whether each of these sequences is graphic. For those that are, draw a graph having the given degree sequence.

- a)** 5, 4, 3, 2, 1, 0   **b)** 6, 5, 4, 3, 2, 1   **c)** 2, 2, 2, 2, 2, 2  
**d)** 3, 3, 3, 2, 2, 2   **e)** 3, 3, 2, 2, 2, 2   **f)** 1, 1, 1, 1, 1, 1  
**g)** 5, 3, 3, 3, 3, 3   **h)** 5, 5, 4, 3, 2, 1

**a)** Since the number of odd-degree vertices has to be even, no graph exists with these degrees. Another reason no such graph exists is that the vertex of degree 0 would have to be isolated but the vertex of degree 5 would have to be adjacent to every other vertex, and these two statements are contradictory.

**b)** Since the number of odd-degree vertices has to be even, no graph exists with these degrees. Another reason no such graph exists is that the degree of a vertex in a simple graph is at most 1 less than the number of vertices.

**c)** A 6-cycle is such a graph. (See picture below.)

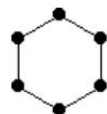
**d)** Since the number of odd-degree vertices has to be even, no graph exists with these degrees.

**e)** A 6-cycle with one of its diagonals added is such a graph. (See picture below.)

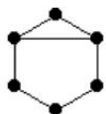
**f)** A graph consisting of three edges with no common vertices is such a graph. (See picture below.)

**g)** The 5-wheel is such a graph. (See picture below.)

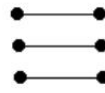
**h)** Each of the vertices of degree 5 is adjacent to all the other vertices. Thus there can be no vertex of degree 1. So no such graph exists.



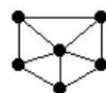
(c)



(e)



(f)



(g)

常见问题：此题错误率较高，尤其c)

66.

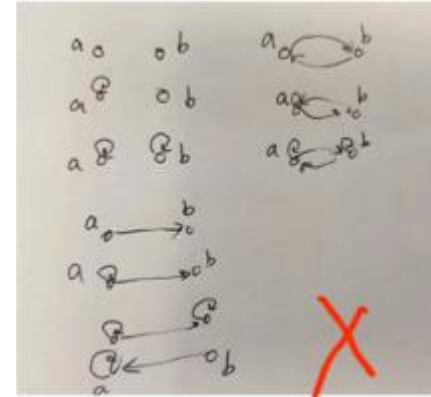
\*66. Show that if  $G$  is a bipartite simple graph with  $v$  vertices and  $e$  edges, then  $e \leq v^2/4$ .

Suppose the parts are of sizes  $k$  and  $v - k$ . Then the maximum number of edges the graph may have is  $k(v - k)$  (an edge between each pair of vertices in different parts). By algebra or calculus, we know that the function  $f(k) = k(v - k)$  achieves its maximum when  $k = v/2$ , giving  $f(k) = v^2/4$ . Thus there are at most  $v^2/4$  edges.

常见问题：多数同学证明过程略简单

74. 4. How many nonisomorphic directed simple graphs are there with  $n$  vertices, when  $n$  is 不同构有向简单图
- a) 2?                      b) 3?                      c) 4?

| TABLE 1 Graph Terminology. |                         |                         |                |
|----------------------------|-------------------------|-------------------------|----------------|
| Type                       | Edges                   | Multiple Edges Allowed? | Loops Allowed? |
| Simple graph               | Undirected              | No                      | No             |
| Multigraph                 | Undirected              | Yes                     | No             |
| Pseudograph                | Undirected              | Yes                     | Yes            |
| Simple directed graph      | Directed                | No                      | No             |
| Directed multigraph        | Directed                | Yes                     | Yes            |
| Mixed graph                | Directed and undirected | Yes                     | Yes            |



a) 2?

Solution : 3

- (1) have no edges from one vertex to the other
- (2) there is an edge from one vertex to the other
- (3) there is an edge from each vertex to the other

常见问题：此题错误率较高，尤其b)，c)，并且部分同学未作答



## 10.5

26. For which values of  $n$  do these graphs have an Euler circuit?

a)  $K_n$  b)  $C_n$  c)  $W_n$  d)  $Q_n$

a) The degrees of the vertices ( $n - 1$ ) are even if and only if  $n$  is odd. Therefore there is an Euler circuit if and only if  $n$  is odd (and greater than 1, of course).

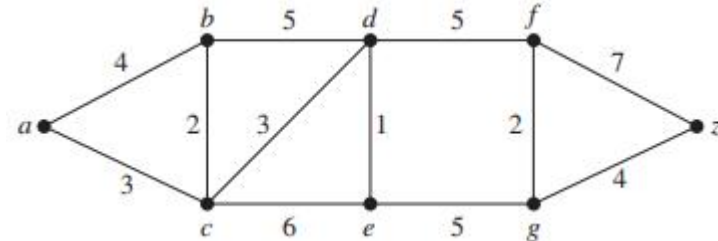
b) For all  $n \geq 3$ , clearly  $C_n$  has an Euler circuit, namely itself.

c) Since the degrees of the vertices around the rim are all odd, no wheel has an Euler circuit.

d) The degrees of the vertices are all  $n$ . Therefore there is an Euler circuit if and only if  $n$  is even (and greater than 0, of course).

10.7 补充题：设 $G$ 是连通平面图， $G$ 的每个面的度数大于等于1 ( $l \geq 3$ )，则 $e \leq l(v-2)/(l-2)$ 。  
用上述性质证明 $K_{3,3}$ 不是平面图。

10.6 Use Dijkstra's Algorithm to find the shortest path length between the vertices  $a$  and  $z$  in this weighted graph.



10.8 23. Find the edge chromatic numbers of

a)  $C_n$ , where  $n \geq 3$ .

b)  $W_n$ , where  $n \geq 3$ .

a) 2 if  $n$  is even, 3 if  $n$  is odd

b)  $n$



- In the questions below fill in the blanks.
- (1) A hypercube  $Q_n$  has  $2^{n-1}$  edges and  $2^n$  vertices.
- (2) The adjacency matrix for  $K_n$  has  $n(n-1)$  1s and  $n$  0s.
- (3) Every Hamilton circuit for  $W_n$  has length  $n+1$ .
- (4) Every Euler circuit for  $K_9$  has length 36.
- (5) List all positive integers  $n$  such that  $C_n$  is bipartite.  $n$  is even
- (6) There are 4 non-isomorphic simple digraphs with 3 vertices and 2 edges.
- (7) The vertex-chromatic number for  $K_7$ ,  $\chi(K_7) = 2$ .
- (1)  $W_n$  has  $2^n$  edges and  $n+1$  vertices.
- (2) The adjacency matrix for  $Q_4$  has 256 entries.
- (3) The length of the longest simple circuit in  $K_{4,10}$  is 40.
- (4) List all positive integers  $n$  such that  $K_n$  has a Hamilton circuit but no Euler circuit.  $n$  even ( $n \geq 2$ ).
- (5) The adjacency matrix for  $K_{m,n}$  has  $m+n$  columns.
- (6) There are 4 non-isomorphic simple undirected graphs with 5 vertices and 3 edges.
- (7) The vertex-chromatic number for  $C_{15}$  is 3.

- (1) Suppose that a planar graph has  $k$  connected components,  $e$  edges, and  $v$  vertices. Also suppose that the plane is divided into  $r$  regions by a planar representation of the graph. Find a formula for  $r$  in terms of  $e$ ,  $v$ , and  $k$ .
- $k$ 个连通部分，增加 $k-1$ 条边，得到连通的平面图。
- $2k-k+1=v-e+r$
- $r=e-v+k+1$
- (2) Suppose that a connected planar graph has six vertices, each of degree four. Into how many regions is the plane divided by a planar representation of this graph?
- 
- $e=6*4/2=12$      $v=6$      $r=e-v+2=8$

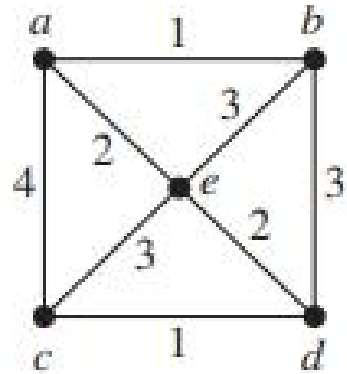
# Trees

- A *tree* is a connected undirected graph with no simple circuits.
  - **Theorem:** There is a unique simple path between any two of its vertices.
- A (not-necessarily-connected) undirected graph without simple circuits is called a *forest*.
- A *rooted tree* is a tree in which one node has been designated the *root*.
  - Parent, child, siblings, ancestors, descendants, leaf, internal node, subtree.

- A rooted tree is called  $n$ -ary if every vertex has no more than  $n$  children.
  - It is called *full* if every internal (non-leaf) vertex has *exactly*  $n$  children.
- A 2-ary tree is called a *binary tree*.
- Any tree with  $n$  nodes has  $e = n - 1$  edges.
  - **Proof:** Consider removing leaves.
- A full  $m$ -ary tree with  $i$  internal nodes has  $n = mi + 1$  nodes, and  $\ell = (m - 1)i + 1$  leaves.
  - **Proof:** There are  $mi$  children of internal nodes, plus the root. And,  $\ell = n - i = (m - 1)i + 1$ .
- Thus, when  $m$  is known and the tree is full, we can compute all four of the values  $e$ ,  $i$ ,  $n$ , and  $\ell$ , given any one of them.
  - $i = (n - 1) / m$  ,  $\ell = [(m - 1)n + 1] / m$
  - $n = mi + 1$  ,  $\ell = (m - 1)i + 1$
  - $n = (m\ell - 1) / (m - 1)$  ,  $i = (\ell - 1) / (m - 1)$

- **Definition:** The *level* of a node is the length of the simple path from the root to the node.
  - The *height* of a tree is maximum node level.
- A rooted  $m$ -ary tree with height  $h$  is called *balanced* if all leaves are at levels  $h$  or  $h-1$ .
- Applications of Trees
- **TREE TRAVERSAL**

- Let  $G$  be a simple graph. A *spanning tree* of  $G$  is a subgraph of  $G$  that is a tree containing every vertex of  $G$ .
- **MINIMUM SPANNING TREES**
- Prim's Algorithm
- Kruskal's Algorithm
- Use Prim's algorithm and Kruskal's Algorithm to find a minimum spanning tree for the given weighted graph.



## Prim's Algorithm

- Procedure *Prim* ( $G$ : weighted connected undirected graph with  $n$  vertices )
- $T :=$  a minimum-weight edge
- For  $i := 1$  to  $n-2$
- begin
  - $e :=$  an edge of minimum weight incident to a vertex in  $T$  and not forming a simple circuit in  $T$  if added to  $T$
  - $T := T$  with  $e$  added
- End{  $T$  is a minimum spanning tree of  $G$  }

## Kruskal's Algorithm

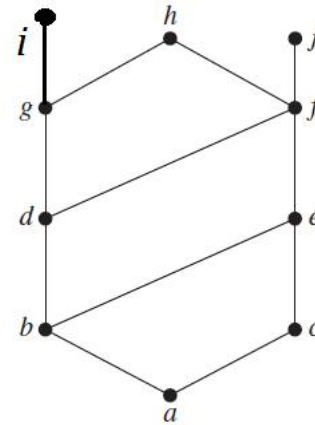
- Procedure Kruskal ( $G$ : weighted connected graph with  $n$  vertices)
- $T := \text{empty graph}$
- for  $i := 1$  to  $n-1$
- begin
  - $e := \text{any edge in } G \text{ with smallest weight that does not form a simple circuit when added to } T$
  - $T := T \text{ with } e \text{ added}$
- end { $T$  is a minimal spanning tree of  $G$ }



## exercise

- In the questions below determine whether the binary relation is: 1) reflexive, 2) symmetric, 3) antisymmetric, 4) transitive.
  - (1) The relation  $R$  on  $\mathbb{Z}$  where  $aRb$  means  $a^2 = b^2$ .
  - (2) The relation  $R$  on the set of all subsets of  $\{1,2,3,4\}$  where  $SRT$  means  $S \subseteq T$ .
  - (3) The relation  $R$  on the set  $\{(a,b) \mid a,b \in \mathbb{Z}\}$  where  $(a,b)R(c,d)$  means  $a = c$  or  $b = d$ .
- Give examples of relations on  $\{1,2,3,4\}$  having the properties specified.
- a) Reflexive, symmetric, and not transitive.
- b) Not reflexive, not symmetric and transitive.
- Use Warshall's algorithm to find the transitive closure of  $R$ .
- a) Find the smallest equivalence relation on the set  $\{1, 2, 3, 4, 5, 6\}$  containing the relation  $\{(1, 2), (2, 3), (4, 5)\}$ .
- b) Find the smallest partial order relation on  $\{1,2,3,4\}$  that contains  $\{(1,1), (3,2), (1,3), (4,1)\}$ .

- Answer these questions for the partial order represented by this Hasse diagram.



- (1) Find all maximal elements.
- (2) Find all minimal elements.
- (3) Is there a greatest element?
- (4) Is there a least element?
- (5) Find all upper bounds of  $\{a, b, c\}$ .
- (6) Find the least upper bound of  $\{b, c, g\}$ , if it exists.
- (7) Find all lower bounds of  $\{f, g, h\}$ .
- (8) Find the greatest lower bound of  $\{j, g, h\}$ , if it exists.
- (9) Use a topological sort to order the elements of the poset .
- (10) Is it a lattice?

## 补充内容：群论、编码见期中复习资料

- Let  $Q$  be the set of rational numbers and define  $a*b=a+b-ab$

- (a) Is  $(Q,*)$  a monoid? Justify your answer.
- (b) If  $(Q,*)$  a monoid, which elements of  $Q$  have an inverse?
- (c) Is  $(Q,*)$  a group? Justify your answer.

| * | e | a | b | c |
|---|---|---|---|---|
| e | e | a | b | c |
| a | a | e | b | c |
| b | b | c | b | c |
| c | c | b | b | c |

- Consider the monoid  $S=\{e,a,b,c\}$  with the following operation table.

- Consider the congruence relation  $R=\{(e,e),(e,a),(a,e),(a,a),(b,b),(b,c),(c,b),(c,c)\}$  on  $S$ .
- (a) Write the operation table of the quotient monoid  $S/R$ .
- (b) Describe the natural homomorphism  $f_R:S\rightarrow S/R$  and give your reason.

- Consider the group  $(Z_6, +_6)$ .

- (a) Write the operation table of the group  $(Z_6, +_6)$ .
- (b) Consider the subgroup  $H = \{ [0], [3] \}$ . Describe the left cosets and right cosets of  $H$  in  $Z_6$ .
- (c) Write the operation table of the quotient group  $Z_6/H$ .
- (d) Determine whether the group  $(Z_2 \times Z_3, )$  and  $(Z_6, +_6)$  are isomorphic.

- Let  $G$  be a group, and let  $H$  and  $K$  be the subgroups of  $G$ . Let the relation  $R$  on  $G$  be defined as follows:

$a R b$  if and only if  $\exists h \in H, \exists k \in K, b = h a k$  Show that  $R$  is an equivalence relation.

- Complete the operational table so that  $*$  is a commutative binary operation.

| $*$ | a | b | c |
|-----|---|---|---|
| a   |   | c | a |
| b   |   |   |   |
| c   |   | b | c |

- $S = \mathbb{Z}$ , with ordinary addition and  $R$  defined by  $a R b$  if and only if  $a \equiv b \pmod{3}$ 
  - a) Prove the relation  $R$  on the semigroup  $(S, +)$  is a congruence relation.
  - b) Write the operation table of the quotient semigroup.
- Write the multiplication table for the group  $\mathbb{Z}_2 \times \mathbb{Z}_3$ . And find all the normal subgroups.

Consider the (2,5) encoding function  $e: B^2 \rightarrow B^5$  defined by

$$e(00) = 00000 \quad e(01) = 01100 \quad e(10) = 10101 \quad e(11) = 11001$$

- Show that the (2,5) encoding function  $e$  is a group code.
- Find the minimum distance of the (2, 5) encoding function  $e$ .
- Determine the number of errors that  $e$  will **detect** and its associated decoding function will **correct**.

Let  $H = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  be a parity check matrix.

- Determine the (2, 5) group code  $e_H: B^2 \rightarrow B^5$ .
- Construct a decoding table relative to a maximum likelihood decoding function associated with  $e_H$ .
- Decode the following words with the decoding table. (1) 11110      (2) 11101
- Compute the syndrome for each coset leader found in (b).
- Compute the syndrome of 11011 and decode it with the syndromes of coset leader.